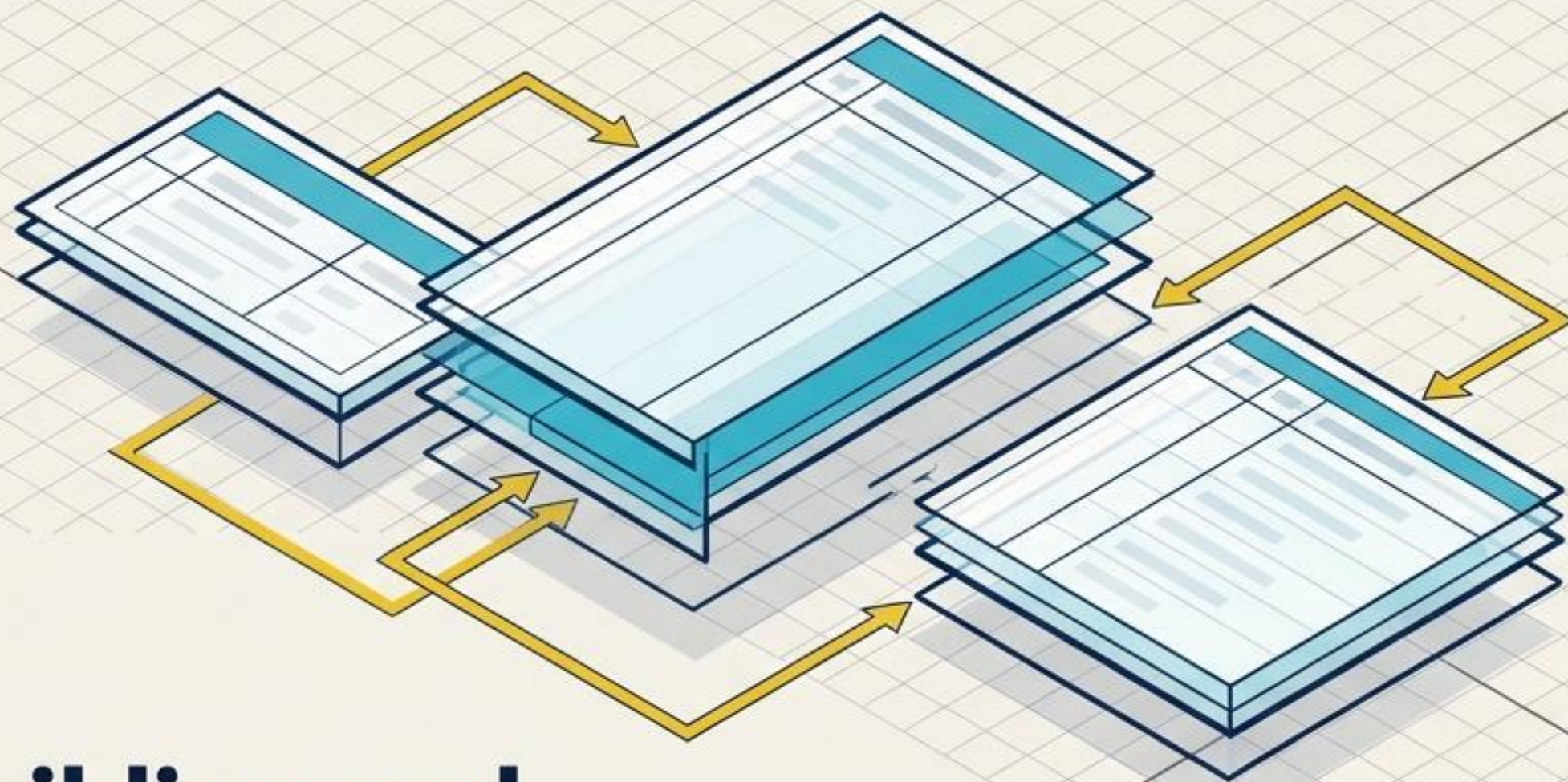


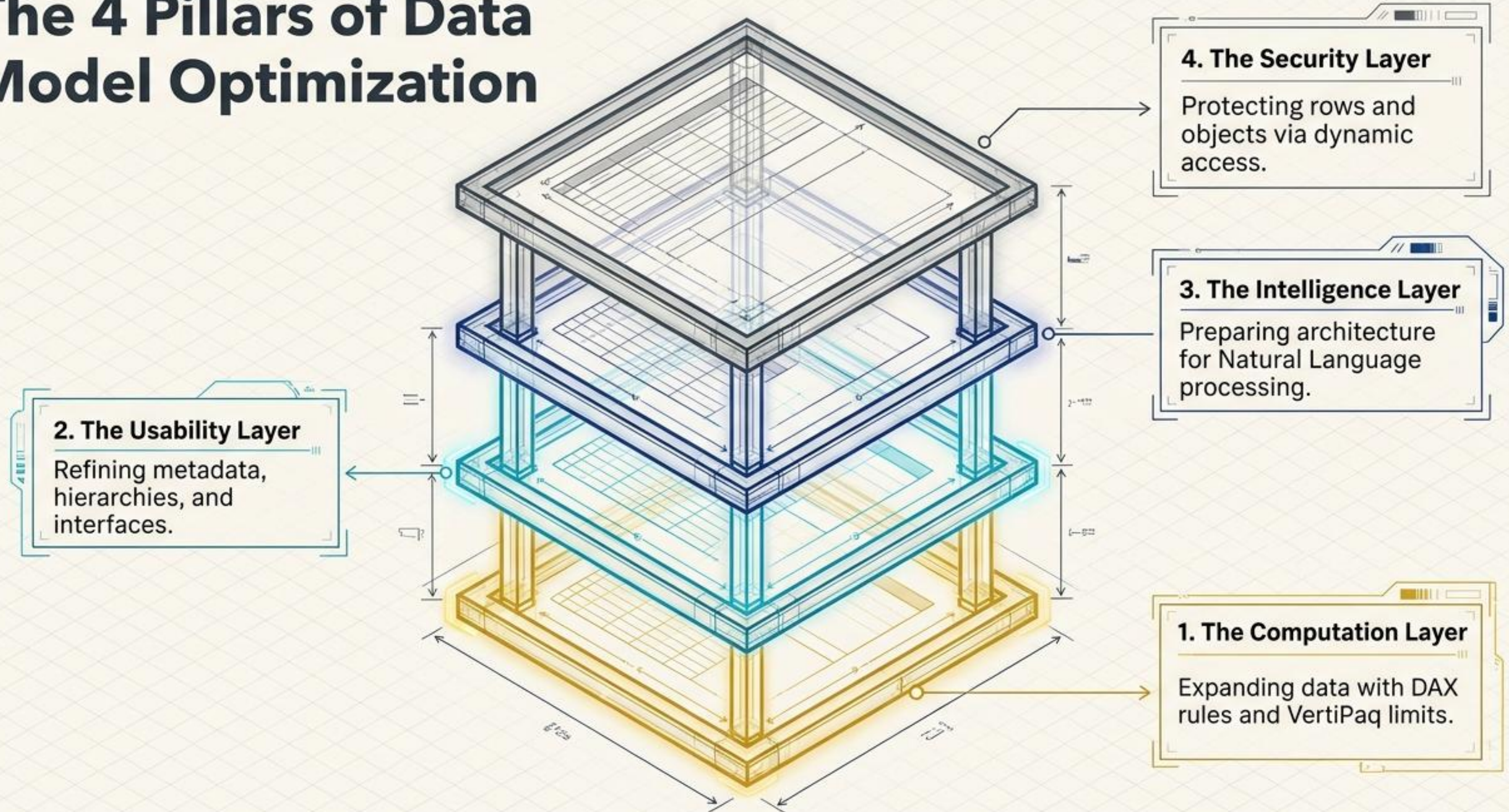


Power BI: Building and Optimization of a Data Model

Best Practices for DAX, Usability, Q&A, and Security

DR ZBIGNIEW GALAR

The 4 Pillars of Data Model Optimization

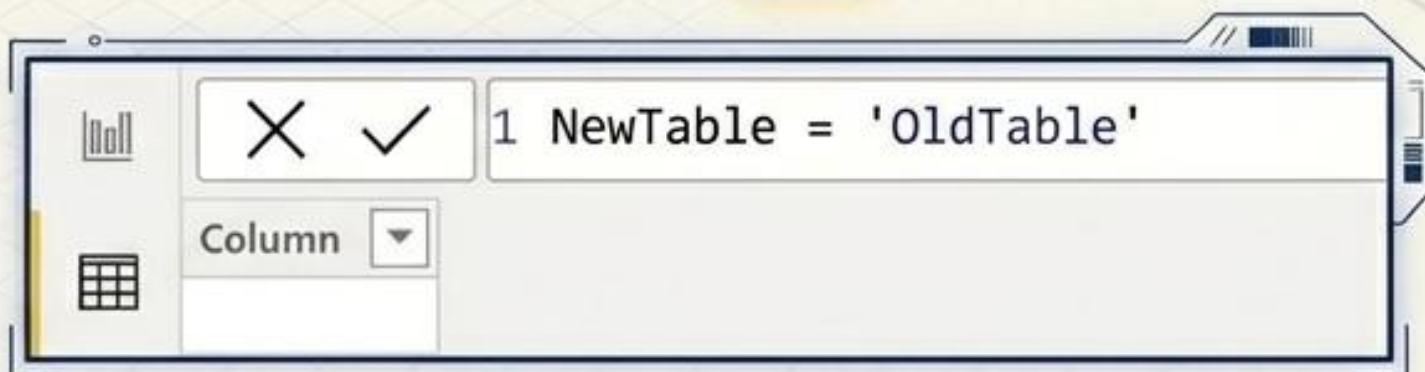


The “Upstream” Principle for Adding Data



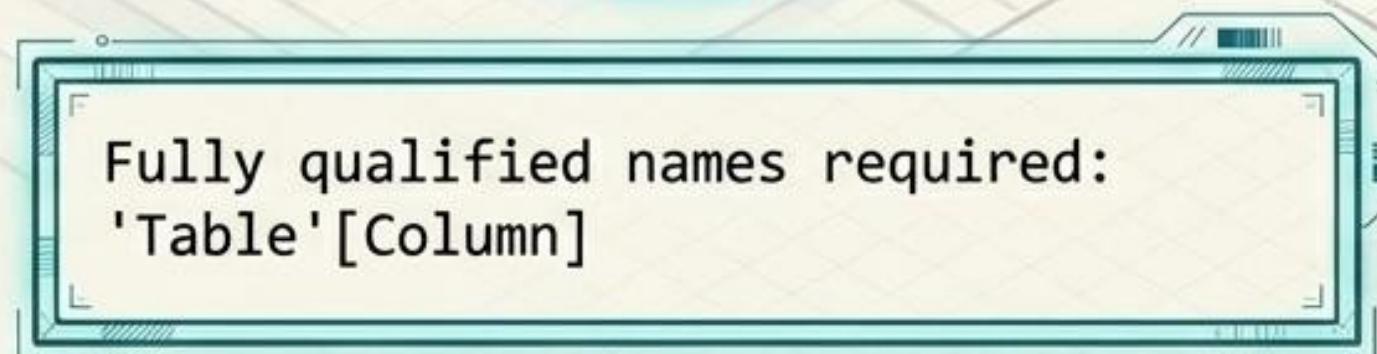
DAX Structural Elements: Tables vs. Columns

Calculated Tables



- **Use Case:** Copying or generating static tables.
- **Warning:** Significant drag on performance; poorly compressed by the VertiPaq engine.

Calculated Columns



- **Use Case:** When results must be used in a slicer/filter, or require row-by-row accuracy.
- **Warning:** Static nature; initiated only when the report opens or model reinitiates.

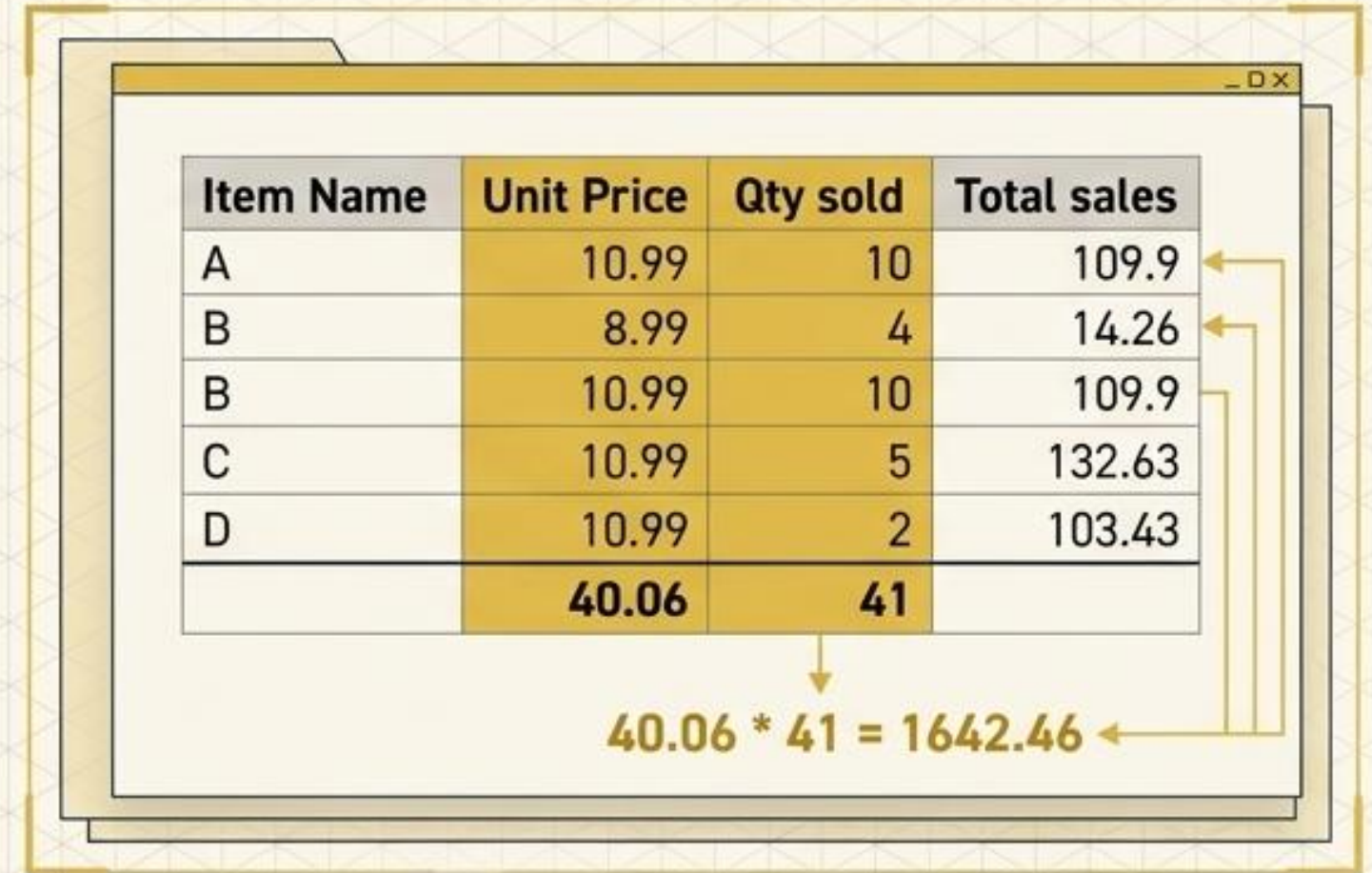
Understanding DAX Row Context

DAX “thinks” primarily in columns. It can also “think” in rows, and when it does, it is iterating.



Item Name	Unit Price	Qty sold	Total sales
A	10.99	10	109.9
B	8.99	4	14.26
B	10.99	10	109.9
C	10.99	5	132.63
D	10.99	2	103.43
			273.63

Row iteration is mandatory for accurate gross totals where individual row states vary.

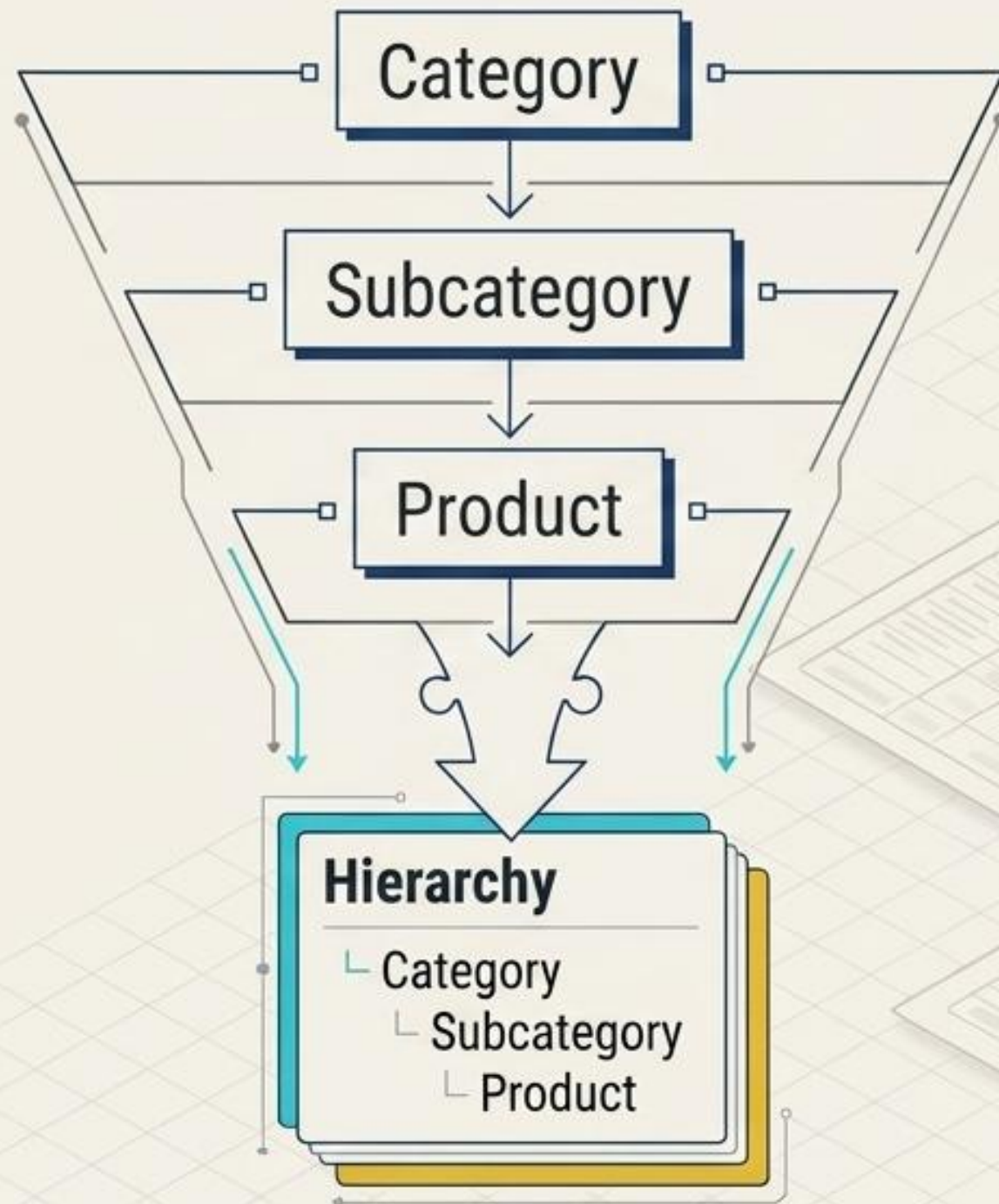


Item Name	Unit Price	Qty sold	Total sales
A	10.99	10	109.9
B	8.99	4	14.26
B	10.99	10	109.9
C	10.99	5	132.63
D	10.99	2	103.43
	40.06	41	

$40.06 * 41 = 1642.46$

Simple column aggregation fails when variables must be evaluated independently per row.

Refinement I: Constructing Hierarchies



Structural Clarity

Signals to the user that fields exist in an established relationship.

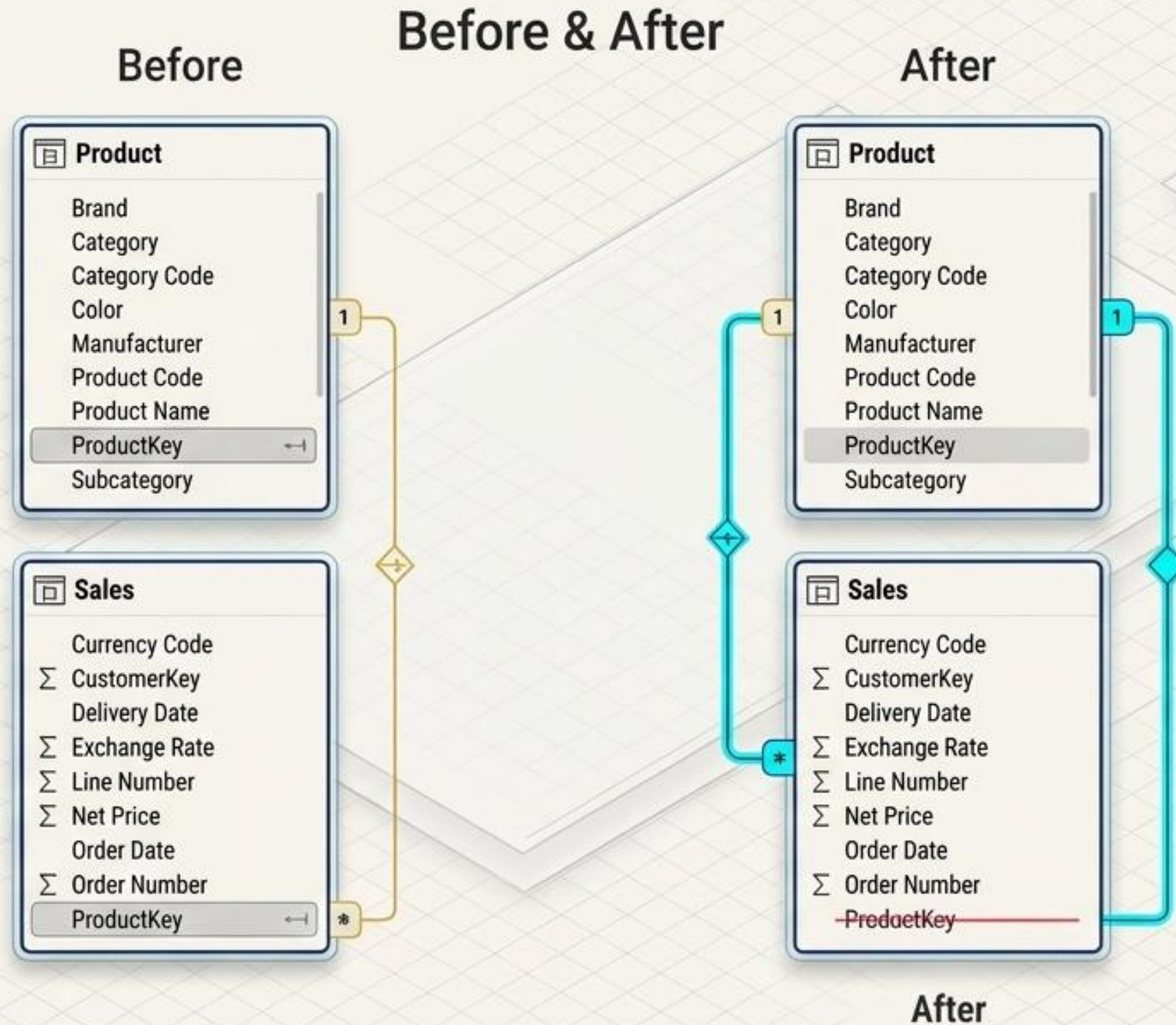
Development Efficiency

Allows report creators to drag all related fields onto a visual simultaneously.

Analytical Depth

Automatically generates a clear, structured drill-down path within visuals.

Refinement II: The Rule of Hiding Fields



The Core Rule

Always hide key fields in the Fact Table.

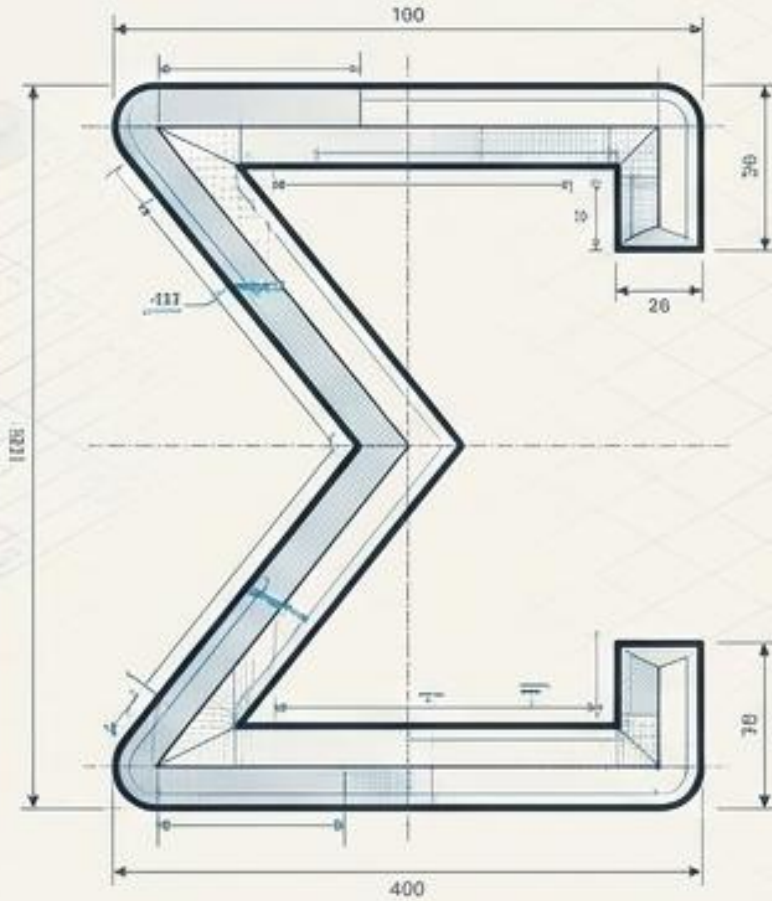
Force consumers to use the labeled fields from the Dimension Table to prevent visual errors.

Helper Fields

Hide index or sort-order fields.

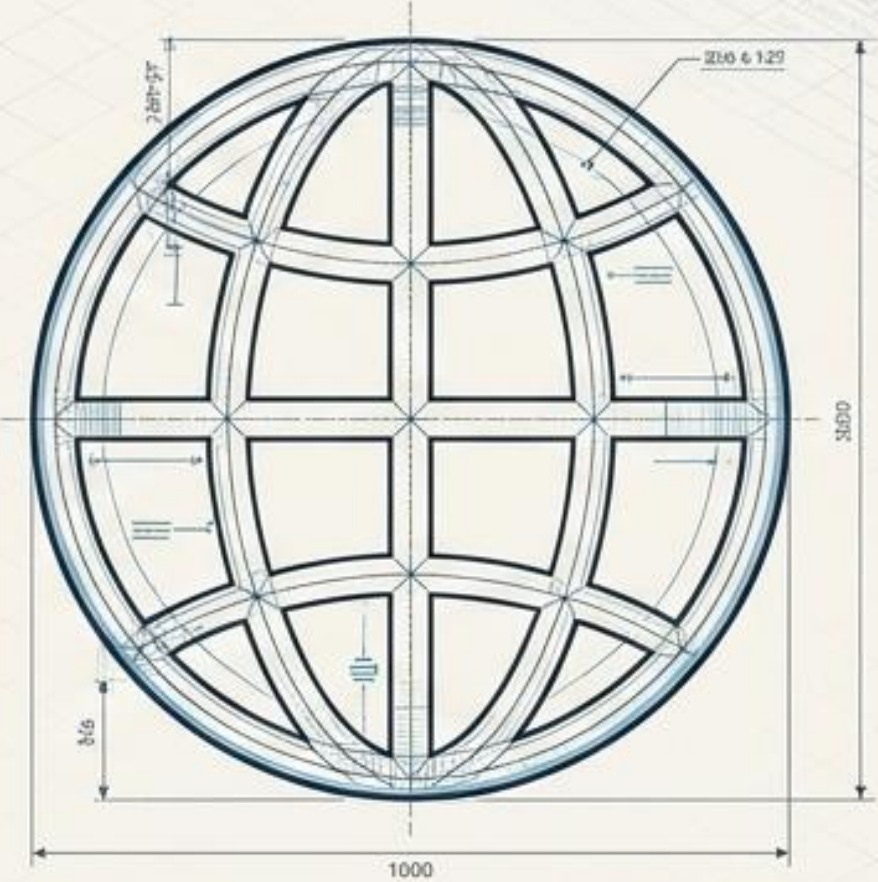
If it provides a backend function but offers no visual analytical value, hide it.

Refinement III: Metadata Tuning



Summarization

Power BI applies default computations (SUM, AVG) to numbers. Removing this ('Don't Summarize') prevents unwanted aggregations on non-additive numbers like IDs.



Categorization

Assigning explicit geographic context or URL functions. Removes mapping ambiguity (e.g., distinguishing 'CA' as California vs. Canada). Zip codes must be formatted as text data types.

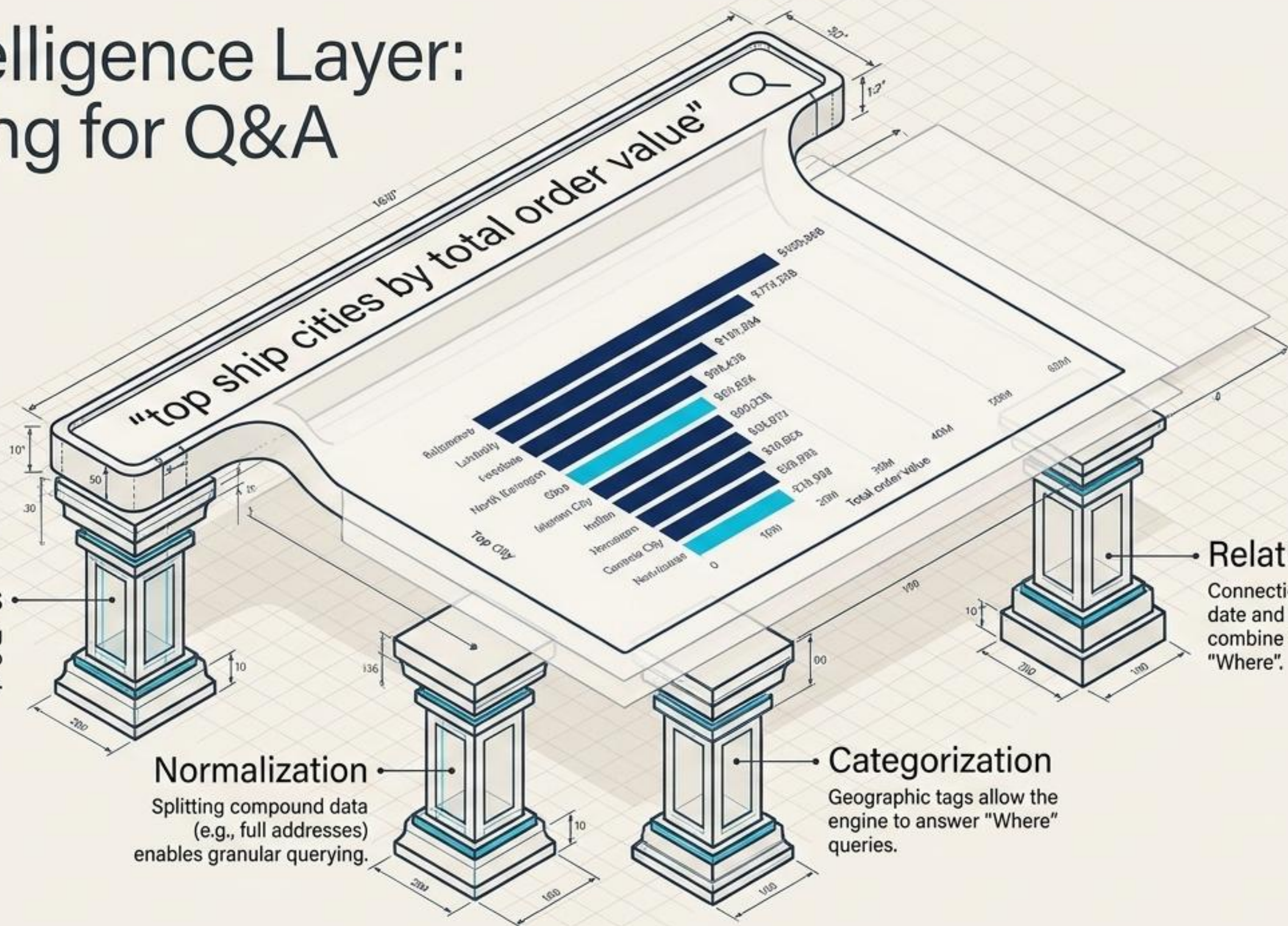
The Intelligence Layer: Preparing for Q&A

Data Types
Proper date formatting
allows the engine to
answer "When" queries.

Normalization
Splitting compound data
(e.g., full addresses)
enables granular querying.

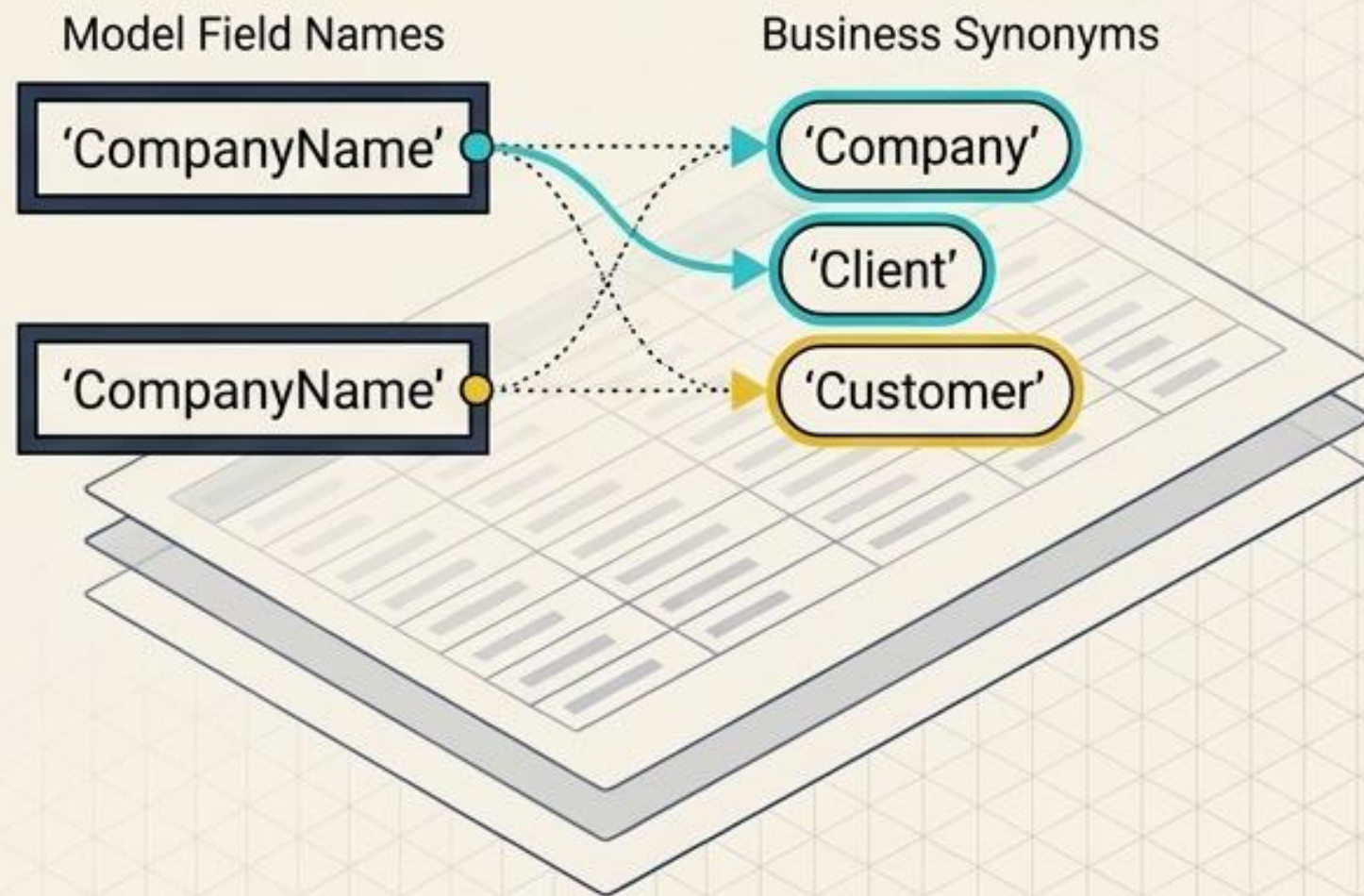
Categorization
Geographic tags allow the
engine to answer "Where"
queries.

Relationships
Connections between
date and location tables
combine "When" and
"Where".



Enhancing Q&A: Teaching the Model

Translation Dictionary



Action 1: Field Synonyms

Map rigid model names to how humans actually speak.

Action 2: Review Questions

Monitor queries people have asked in the Service and fix misunderstandings to train the engine.

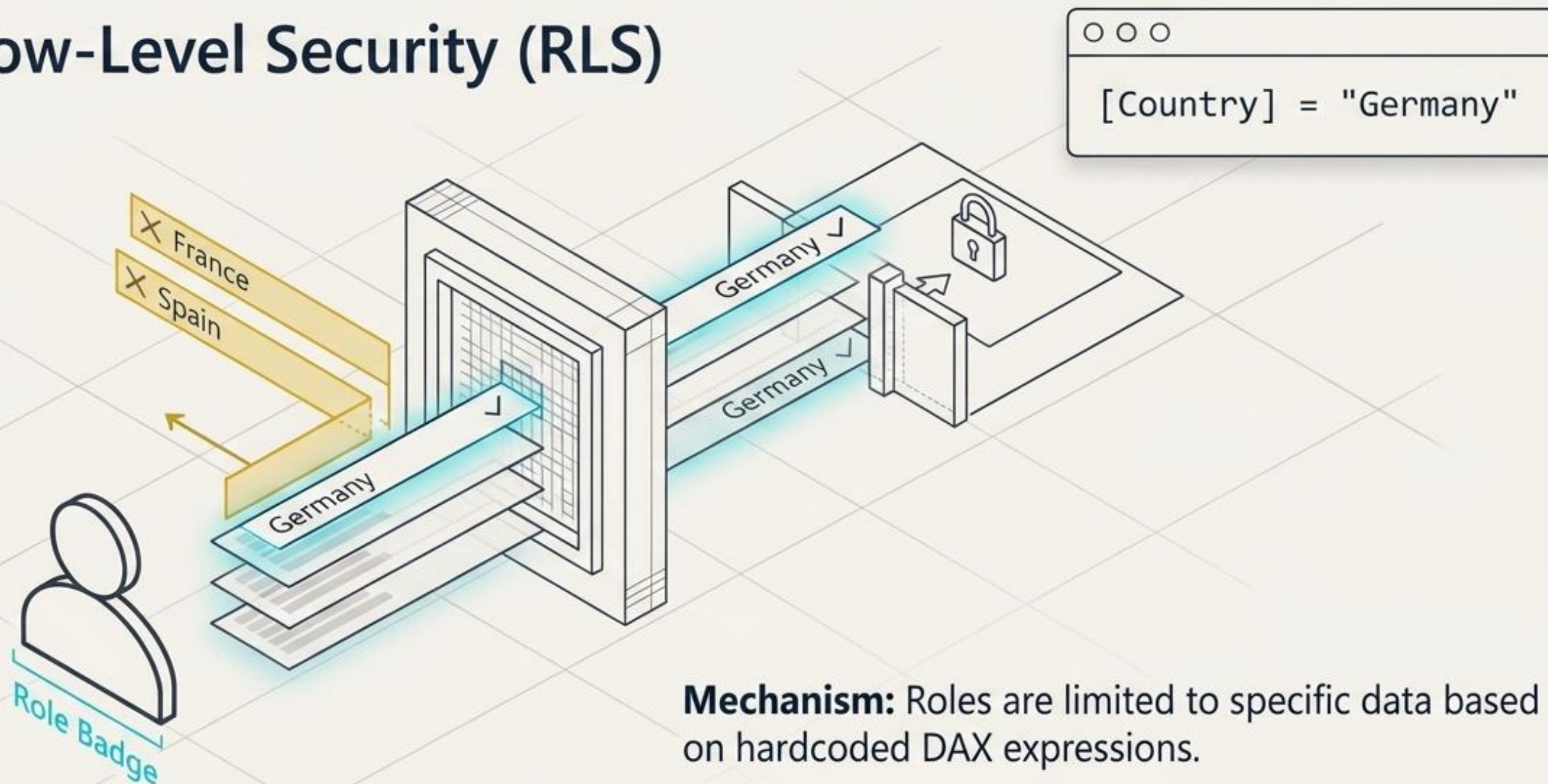
Action 3: Suggest Questions

Create 'seed' questions to help users explore a new dataset immediately.

The Multilayer Security Architecture



Static Row-Level Security (RLS)

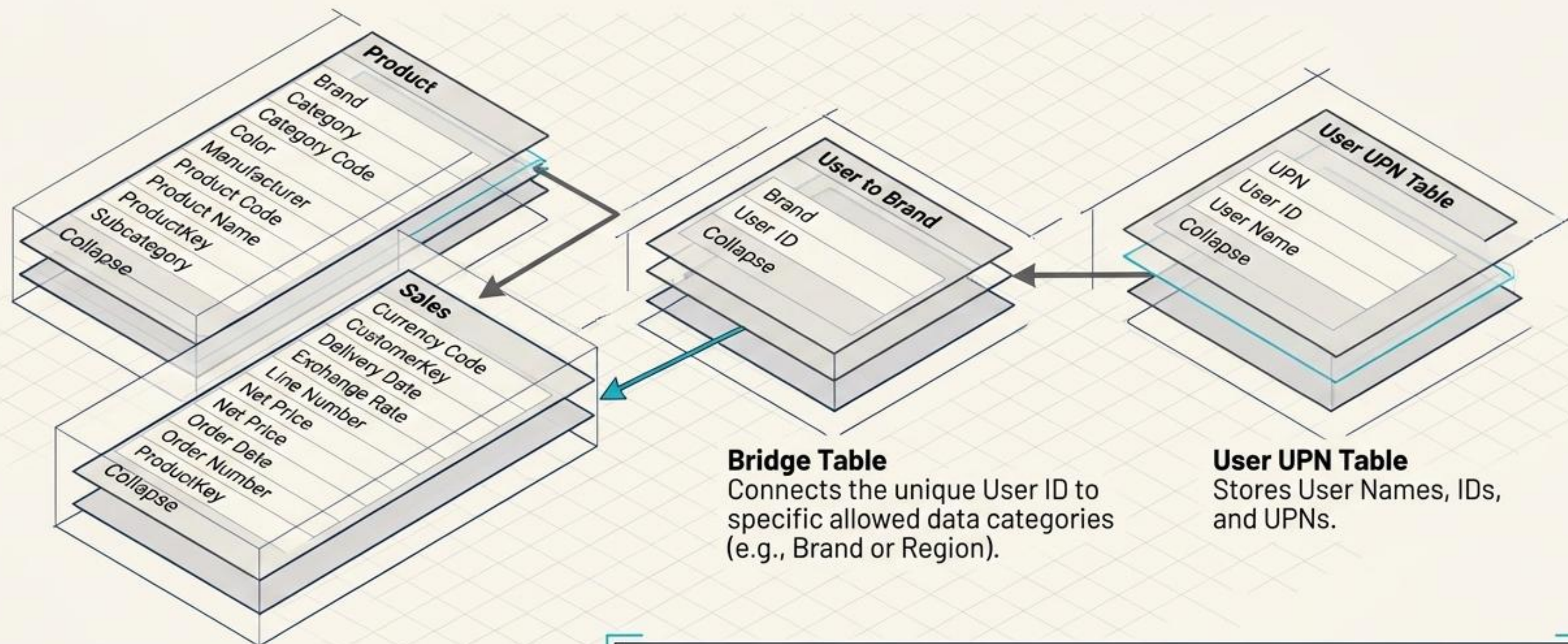


⚠ Now viewing as: Germany Account Manager

Mechanism: Roles are limited to specific data based on hardcoded DAX expressions.

Validation: Security boundaries must be tested in Desktop using the 'View as' feature to confirm visuals properly restrict data prior to Service publication.

Dynamic RLS: Conceptual Flow



Fact/Dim Tables
The actual report data.

The Concept: Filter direction must flow downstream from the User Table, through the Bridge Table, to automatically filter the Fact Data based on who is logged in.

Dynamic RLS: Implementation Code



Table filter DAX expression

```
1 / The Engine: USERPRINCIPALNAME()  
2 [User ID] = USERPRINCIPALNAME()  
3 |
```

Implementation:

Create a single role with a DAX expression explicitly declaring the user's access boundaries based on their login.

Result:

Scalable security. Hundreds of users can be managed via one role, with the UPN dynamically determining their unique filter context.

The Optimization Master Checklist

	Performance	✓ Are DAX calculation loads pushed upstream where possible? ✓ Are calculated tables avoided to preserve VertiPaq compression?
	Usability	✓ Are Fact table keys hidden to prevent confusion? ✓ Are hierarchies built for intuitive drill-downs? ✓ Are default summarization behaviors verified?
	AI Readiness	✓ Are postal codes properly formatted as text? ✓ Is all geography mapped and categorized? ✓ Are business synonyms defined in Q&A setup?
	Security	✓ Is row-level access validated using the 'View As' feature? ✓ Is Dynamic RLS architecture mapped to USERPRINCIPALNAME()?